THE UNIVERSITY OF AUCKLAND

SEMESTER TWO, 2024
Campus: City

MATHEMATICS

Real Analysis

(Time allowed: TWO hours)

NOTE: Please answer all **four questions**. Each question is worth 20 marks, for a total of 80 marks.

Show all working and state clearly any results that you use in your answers.

Calculators are **not permitted** in this examination.

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1. (a) Use the ε-N definition for the limit of a sequence to prove that

$$\lim_{n\to\infty} \frac{3n^2+n-1}{n^2} = 3.$$

(b) Consider the sequence

$$a_n := \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} \quad (n \text{ terms}).$$

Show that (a_n) converges.

Hint: First show that (a_n) is increasing by considering the difference $a_{n+1} - a_n$.

(c) Consider a sequence of real numbers of the form

$$(x_1, -x_1, x_2, -x_2, x_3, -x_3, \ldots).$$

Find all possible limits that such a sequence could have.

Hint: Consider subsequences.

(d) Let $f : [a, b] \to \mathbb{R}$ be continuous, and suppose that f is not bounded above, i.e., there exists a sequence (x_n) of points in $[a, b]$ with

$$\lim_{n\to\infty} f(x_n) = \infty.$$

Show that this implies f is not continuous at some point $c \in [a, b]$.

Hint: Use Bolzano-Weierstrass and the sequential criterion for continuity at a point.

2. Let $f : I \to \mathbb{R}$ be differentiable on an interval I which contains at least two points.

(a) Show that if the derivative f' is bounded, then f is Lipschitz continuous.

Hint: Use the mean value theorem.

(b) Show that if f is Lipschitz continuous, then f' is bounded.

Hint: Consider the size of the Newton quotient.

(c) Suppose that $I = [a, b]$ and f' is continuous. Explain why f is Lipschitz continuous.

(d) Give an example of a differentiable function $g : (0, 1] \to \mathbb{R}$ which is not Lipschitz continuous.

3. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) := \int_0^x e^{-t^2}\, dt.$$

(a) Calculate the derivative of f, and explain why f is infinitely differentiable.

(b) Show that f is an odd function, i.e., $f(-x) = -f(x)$, for all $x \in \mathbb{R}$.

(c) Calculate the quadratic Taylor polynomial $T_2(f, x)$ with centre 0 for $f(x)$.

(d) Find the power series with centre 0 for $f(x)$.

Hint: The exponential is defined by the power series $e^z := \sum_{n=0}^{\infty} \frac{z^n}{n!}$.

(e) What is the radius of convergence of the above power series for $f(x)$?

4. Let $f : [a, b] \to \mathbb{R}$ be a bounded function, and $P = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$, with

$$a = x_0 < x_1 < \cdots < x_n = b.$$

(a) Define the **upper** and **lower sums** $U(f, P)$ and $L(f, P)$ of f for the partition P.

(b) Define the **upper** and **lower integrals** of f over $[a, b]$ in terms of upper and lower sums.

(c) Give conditions for f to be Riemann integrable over $[a, b]$ in terms of

(i) the upper and lower integrals,

(ii) the upper and lower sums.

(d) Show that if $f \geq 0$, then $U(f, P) \geq 0$, and use this to show that $\int_a^b f \geq 0$, when f is Riemann integrable.

Hint: Consider the upper integral of f.

(e) For a choice of points $x_j^* \in [x_{j-1}, x_j]$, $j = 1, \ldots, n$, define a "middle sum" by

$$M(f, P) := \sum_{j=1}^{n} f(x_j^*)(x_j - x_{j-1}).$$

Now let $f : [0, 1] \to \mathbb{R}$ be the function defined by

$$f(x) := \begin{cases} 1, & x \in \mathbb{Q}; \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

(i) Show that $0 = L(f, P) \leq M(f, P) \leq U(f, P) = 1$.

(ii) Fix $0 < c < 1$. Show that there is a partition P of $[0, 1]$ and a choice of points x_j^* with $M(f, P) = c$.
Hint: Consider a partition with a small number of points.